Control Statements in Python (if statement)

With Practical

Outcomes of this Lecture

- Sequential Execution
- Need of Control Statements
- Categories of Control Statements
- Importance of Indentation
- Implementation of if statement

What is the need of Control Statements?

- By default, statements in the program are executed sequentially from the first to the last, is called **“sequential execution”.** It is suitable to develop simple programs.
- To develop critical programs, it may be required to change the flow of execution.
- Flow control describes the order in which statements will be executed at runtime.

Use of Control Statements

- Control statements in Python are used to **control the flow of execution** in a program.
- Instead of executing line-by-line sequentially, they **help us make decisions, repeat tasks, or jump between parts of code.**

Categories

Control Statements

- Selection Statements
- Transfer Statements
- Iterative Statements

Note: Selection Statements are also called **Branching Statements** or **Decision Making**

Selection Statements

- Used to **make decisions based on condition(s)**
- A block of statement or a block of statements will be executed if a certain expression (condition) is true.
- If condition is false another set of statements will be executed.

Selection Statements Contd..

- The **Comparison/Relational** and **logical operators** are useful in controlling flow of the program.
- Ex: Printing “Good Morning” or “Good Evening” **based on time of the day.**

Types of Control Statements

Selection Statements

- If statement
- If-else statement
- If-elif-else
- Match-case

Note:

The match-case statement was introduced in **2021** with the release of **Python 3.10**

If statement

- If statement is used to execute single statement or block of statements depending on whether the given condition is True or False.
- **Real-life usage:** A website checking if a user has entered a password of at least 8 characters before allowing them to sign up.

Syntax: If statement

- Syntax: **if condition:**

  statement-1

  statement-2

  ……..

  statement-n

If condition is true then statements will be executed.

Indentation

- After the **: symbol** start a block with **increased indent**. One or more statements written with the same level of indent will be executed if the given condition evaluates to True.
- To **end the block**, **decrease the indentation**. Subsequent statements after the block will be executed out of the if condition.

Check your knowledge

- What is Sequential Execution?
- What is the need of Control Statements?
- What are the categories of control statements?
- What is the use of Indentation in Python?
- How to write python program to find largest of three given numbers?